Root systems

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May 15, 2026

Chapter 1

Root pairings

1.1 Definitions

Recall that a *perfect pairing* is a quadruple $(R, M, N, \mathcal{L})$ where R is a ring, M and N are R-modules, and $\mathcal{L} : M \times N \to R$ is a bilinear map such that the induced maps $M \to \mathrm{Hom}_R(N, R)$ and $N \to \mathrm{Hom}_R(M, R)$ are isomorphisms of R-modules.

If I is a set, denote by $\mathrm{Perm}(I)$ the group of permutations of I.

Definition 1. A *root pairing* is a tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ where $(R, M, N, \mathcal{L})$ is a perfect pairing, I is a set, $\alpha : I \to M, i \mapsto \alpha_i$, $\alpha^\vee : I \to N, i \mapsto \alpha_i^\vee$, and $s : I \to \mathrm{Perm}(I), i \mapsto s_i$, with the conditions that

1. $\alpha, \alpha^\vee$ are injective,
2. For all $i \in I$, $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$,
3. For all $i, j \in I$, we have

$$\begin{aligned} \alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee)\alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee)\alpha_i^\vee. \end{aligned}$$

1.2 Root pairing of type A

1.2.1 Construction

Consider the perfect pairing $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$ where $(\cdot, \cdot)$ is the canonical pairing between $\mathbb{Z}^n$ and its algebraic dual $(\mathbb{Z}^n)^* := \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$, i.e. the dot product.

We denote by $e_1, ..., e_n$ the standard basis of $\mathbb{Z}^n$ and by $e_1^*, ..., e_n^*$ the dual basis of $(\mathbb{Z}^n)^*$.

Write $[1, n] = \{1, ..., n\}$. An *interval* in $[1, n]$ is a subset of the form $[i, j] \overset{\text{denotes}}{=} \{i, i+1, ..., j\}$ for some $1 \leq i \leq j \leq n$. We denote by I_n the set of intervals in $[1, n]$. Write $[i] := [i, i]$ for $1 \leq i \leq n$.

A *signed interval* in $[1, n]$ is an element of $I_n \times \{1, -1\}$. We denote by I_n^+ the set of signed intervals of the form $(J, 1)$ and by I_n^- the set of signed intervals of the form $(J, -1)$. By abuse of notation, we identify I_n with I_n^+ and write J for the signed interval $(J, 1)$ for all $J \in I_n$. If $J \in I_n$ then we write $-J$ for the signed interval $(J, -1)$.

Let $\mathcal{A}_n = (a_{i,j})$ be the $n \times n$ integer matrix defined by

i) $a_{i,i} = 2$ for all $i \in [1, n]$,
ii) $a_{i,j} = -1$ if $|i - j| = 1$, and
iii) $a_{i,j} = 0$ otherwise.

The matrix $\mathcal{A}_n$ is often called a *Cartan matrix of type* A_n. In the rest of this subsection we write $A = \mathcal{A}_n$.

Lemma 2. *For any* $n \in \mathbb{Z}_{\geq 1}$, $\det(\mathcal{A}_n) = n + 1$. *In particular,* $\mathcal{A}_n$ *is invertible.*

Proof. By a direct computation, $\det(\mathcal{A}_1) = 2$ and $\det(\mathcal{A}_2) = 3$. Now fix $n \geq 3$, and suppose that the claim holds for all $k < n$. Using the Laplace expansion along the first row, we have

$$\begin{aligned}\det(\mathcal{A}_n) &= 2\det(\mathcal{A}_{n-1}) - \det(\mathcal{A}_{n-2}) \\ &= 2n - (n-1) = n + 1.\end{aligned}$$

The statement follows. □

Corollary 3. *The set* $\{\mathcal{A}_n e_k : 1 \leq k \leq n\}$ *is linearly independent over* $\mathbb{Q}$.

Proof. If $\sum_{k=1}^{n} c_k \mathcal{A}_n e_k = 0$ for some $c_1, ..., c_n \in \mathbb{Q}$, then $\mathcal{A}_n(\sum_{k=1}^{n} c_k e_k) = 0$. Left-multiplying both sides by $\mathcal{A}_n^{-1}$, we have $\sum_{k=1}^{n} c_k e_k = 0$. Since $\{e_k : 1 \leq k \leq n\}$ is a basis of $\mathbb{Z}^n$, we have $c_k = 0$ for all k. □

Define

$$\alpha : I_n \times \{\pm 1\} \to \mathbb{Z}^n, \qquad \forall J \in I_n,\ \ \alpha_{\pm J} = \pm(\sum_{j \in J} Ae_j), \tag{1.1}$$

$$\alpha^\vee : I_n \times \{\pm 1\} \to (\mathbb{Z}^n)^*, \qquad \forall J \in I_n,\ \ \alpha^\vee_{\pm J} = \pm(\sum_{j \in J} e_j^*). \tag{1.2}$$

Lemma 4. *The maps* $\alpha : I_n \times \{\pm 1\} \to \mathbb{Z}^n$ *and* $\alpha^\vee : I_n \times \{\pm 1\} \to (\mathbb{Z}^n)^*$ *are injective.*

Proof. Let $J, K \in I_n \times \{\pm 1\}$. Write $J = \varepsilon J_0$ and $K = \delta K_0$, where $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Suppose first that $\alpha_J = \alpha_K$. Then

$$\sum_{r=1}^{n} (\varepsilon \mathbf{1}_{r \in J_0} - \delta \mathbf{1}_{r \in K_0}) Ae_r = 0.$$

Since the vectors $Ae_1, \dots, Ae_n$ are linearly independent, each coefficient is zero. Thus

$$\varepsilon \mathbf{1}_{r \in J_0} = \delta \mathbf{1}_{r \in K_0} \qquad \text{for all } r \in [1, n].$$

It follows that $r \in J_0$ if and only if $r \in K_0$, so $J_0 = K_0$; since these intervals are nonempty, the same equality of coefficients also gives $\varepsilon = \delta$. Hence $J = K$.

The proof for $\alpha^\vee$ is identical, using the linear independence of the dual basis $e_1^*, \dots, e_n^*$. Indeed, if $\alpha^\vee_J = \alpha^\vee_K$, then

$$\sum_{r=1}^{n} (\varepsilon \mathbf{1}_{r \in J_0} - \delta \mathbf{1}_{r \in K_0}) e_r^* = 0,$$

and the same coefficient comparison gives $J_0 = K_0$ and $\varepsilon = \delta$. □

Lemma 5. *For every* $J \in I_n \times \{\pm 1\}$, $\mathcal{L}(\alpha_J, \alpha^\vee_J) = 2$.

Proof. Write $J = \varepsilon J_0$, where $J_0 = [p, q]$ and $\varepsilon \in \{\pm 1\}$. Since $\varepsilon^2 = 1$,

$$\begin{aligned}\mathcal{L}(\alpha_J, \alpha_J^\vee) &= \sum_{u=p}^{q} \sum_{v=p}^{q} e_v^*(Ae_u) \\ &= \sum_{u=p}^{q} \sum_{v=p}^{q} a_{v,u}.\end{aligned}$$

If $m = q - p + 1$ is the length of J_0, this sum has m diagonal terms, each equal to 2, and $2(m-1)$ ordered adjacent off-diagonal terms, each equal to -1. All other terms are zero. Hence

$$\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2m - 2(m-1) = 2.$$

$\square$

We first define $s_J(K)$ for $J, K \in I_n$. Set $L = (J \cup K) \backslash (J \cap K)$. It is clear that J, K, L satisfy exactly one of the following four conditions.

R1. $L = \emptyset$.

R2. $L \in I_n$ and $K \subset J$,

R3. $L \in I_n$ and $K \not\subseteq J$,

R4. $L \notin I_n \cup \{\emptyset\}$.

Then,

$$s_J(K) = \begin{cases} -K & \text{in case } \mathbf{R1}, \\ -L & \text{in case } \mathbf{R2}, \\ L & \text{in case } \mathbf{R3}, \\ K & \text{in case } \mathbf{R4}. \end{cases} \tag{1.3}$$

If $J = [i, j]$ and $K = [k, l]$, then (1.3) can be rewritten as

$$s_{[i,j]}[k, l] = \begin{cases} -[k, l] & \text{if } (i, j) = (k, l) \\ -[l + 1, j] & \text{if } i = k, j > l \\ [j + 1, l] & \text{if } i = k, j < l \\ -[i, k - 1] & \text{if } i < k, j = l \\ [k, i - 1] & \text{if } i > k, j = l \\ [k, j] & \text{if } i = l + 1 \\ [i, l] & \text{if } k = j + 1 \\ [k, l] & \text{otherwise.} \end{cases} \tag{1.4}$$

We extend s_J to $I_n \times \{\pm 1\}$ by setting $s_J(-K) = -s_J(K)$ for all $K \in I_n$. We extend s to $I_n \times \{\pm 1\}$ by setting $s_{-J} = s_J$ for all $J \in I_n$.

Lemma 6. *For all $J \in I_n \times \{\pm 1\}$, we have $s_J \in \mathrm{Perm}(I_n \times \{\pm 1\})$.*

Proof. Fix $J \in I_n \times \{\pm 1\}$. We prove that $s_J^2 = \mathrm{id}$ by a straightforward case-by-case check using the definition of s_J and the four cases above. Then $s_J^{-1} = s_J$, and our claim is proved. $\square$

Lemma 7. *For all $J, K \in I_n \times \{\pm 1\}$,*

$$\alpha_{s_J(K)} = \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee)\alpha_J.$$

Proof. First assume that $J = [i,j]$ and $K = [k,l]$ are positive intervals. Let $c = \mathcal{L}(\alpha_K, \alpha_J^\vee)$. For $u \in [1,n]$,

$$\sum_{r=k}^{l} a_{u,r} = \mathbf{1}_{u=k} + \mathbf{1}_{u=l} - \mathbf{1}_{u=k-1} - \mathbf{1}_{u=l+1}.$$

Summing this identity over $u = i, \dots, j$ gives

$$c = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{j,k-1},$$

where δ denotes the Kronecker delta. Comparing this value with (1.4), we get the following cases:

condition	c	$\alpha_K - c\alpha_J$
$(i,j) = (k,l)$	2	$-\alpha_{[k,l]}$
$i = k,\ j > l$	1	$-\alpha_{[l+1,j]}$
$i = k,\ j < l$	1	$\alpha_{[j+1,l]}$
$i < k,\ j = l$	1	$-\alpha_{[i,k-1]}$
$i > k,\ j = l$	1	$\alpha_{[k,i-1]}$
$i = l+1$	-1	$\alpha_{[k,j]}$
$k = j+1$	-1	$\alpha_{[i,l]}$
otherwise	0	$\alpha_{[k,l]}$.

Each entry in the last column is exactly $\alpha_{s_J(K)}$ by (1.4). Thus the desired identity holds for positive intervals.

Now write arbitrary signed intervals as $J = \varepsilon J_0$ and $K = \delta K_0$, with $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Since $s_{\varepsilon J_0}(\delta K_0) = \delta s_{J_0}(K_0)$ and $\mathcal{L}(\alpha_K, \alpha_J^\vee) = \delta\varepsilon\mathcal{L}(\alpha_{K_0}, \alpha_{J_0}^\vee)$, the positive case gives

$$\begin{aligned}\alpha_{s_J(K)} &= \delta\alpha_{s_{J_0}(K_0)} \\ &= \delta(\alpha_{K_0} - \mathcal{L}(\alpha_{K_0}, \alpha_{J_0}^\vee)\alpha_{J_0}) \\ &= \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee)\alpha_J.\end{aligned}$$

□

Lemma 8. *For all $J, K \in I_n \times \{\pm 1\}$,*

$$\alpha^\vee_{s_J(K)} = \alpha_K^\vee - \mathcal{L}(\alpha_J, \alpha_K^\vee)\alpha_J^\vee.$$

Proof. First assume that $J = [i,j]$ and $K = [k,l]$ are positive intervals, and set $c = \mathcal{L}(\alpha_J, \alpha_K^\vee)$. Since A is symmetric, the endpoint computation in the proof of Lemma 7 gives the same value

$$c = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{j,k-1}.$$

The case table in that proof used only additivity of sums over intervals. Replacing each Ae_r by e_r^* in the last column gives

$$\alpha^\vee_{s_J(K)} = \alpha_K^\vee - c\alpha_J^\vee$$

for positive intervals.

For general signed intervals, write $J = \varepsilon J_0$ and $K = \delta K_0$, with $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Using $s_{\varepsilon J_0}(\delta K_0) = \delta s_{J_0}(K_0)$ and $\mathcal{L}(\alpha_J, \alpha_K^\vee) = \varepsilon\delta\mathcal{L}(\alpha_{J_0}, \alpha_{K_0}^\vee)$, the positive case gives

$$\begin{aligned} \alpha^\vee_{s_J(K)} &= \delta\alpha^\vee_{s_{J_0}(K_0)} \\ &= \delta(\alpha^\vee_{K_0} - \mathcal{L}(\alpha_{J_0}, \alpha^\vee_{K_0})\alpha^\vee_{J_0}) \\ &= \alpha^\vee_K - \mathcal{L}(\alpha_J, \alpha^\vee_K)\alpha^\vee_J. \end{aligned}$$

□

Theorem 9. *Fix $n \in \mathbb{Z}_{>0}$, $R = \mathbb{Z}$, $M = \mathbb{Z}^n$ and $N = (\mathbb{Z}^n)^*$. Let $\mathcal{L}$ be the canonical pairing, let $I = I_n \times \{\pm 1\}$ be the set of signed intervals, let α and $\alpha^\vee$ be as in (1.1) and (1.2) respectively, and let s be as in (1.3). Then the 8-tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ is a root pairing.*

Proof. The quadruple $(\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, \mathcal{L})$ is the standard perfect pairing between a finite free $\mathbb{Z}$-module and its dual. It remains to verify the three extra conditions in Definition 1.

The maps α and $\alpha^\vee$ are injective by Lemma 4. For every signed interval $J \in I$, Lemma 5 gives $\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2$. For every $J \in I$, Lemma 6 gives $s_J \in \mathrm{Perm}(I)$. Finally, for all $J, K \in I$, Lemmas 7 and 8 give

$$\alpha_{s_J(K)} = \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee)\alpha_J \quad \text{and} \quad \alpha^\vee_{s_J(K)} = \alpha^\vee_K - \mathcal{L}(\alpha_J, \alpha^\vee_K)\alpha^\vee_J.$$

These are precisely the axioms in Definition 1, so the stated 8-tuple is a root pairing. □

1.2.2 Properties

1. Crystallographic root pairing
2. Reduced root pairing
3. Finite root pairing
4. Irreducible root pairing
5. Base of a root pairing
6. Cartan matrix of a root pairing
7. Type of a reduced, finite, irreducible crystallographic root pairing